# CRIMENEXUS v3.0 • SYSTEM CAPABILITY APPENDIX

## Frontline Police UX (10) · Advanced AI/ML (10) · Operational Excellence (10)

| # | Capability / Module | Category | Operational Police Value | Verification Proof |
|---|---|---|---|---|
| 1 | Interactive Onboarding Tour | Frontline UX | Zero-training UI guidance for IOs on Cytoscape/CDR | react-joyride modal test |
| 2 | One-Click Court Dossier PDF | Frontline UX | Auto-compiles 65B BSA cert + graph exhibits in 4s | ReportLab test passed |
| 3 | District Most Wanted Watchlist | Frontline UX | Ranked threat leaderboard with Court XAI breakdown | API /threats/most-wanted |
| 4 | Visual Case History Timeline | Frontline UX | Chronological audit events from FIR to chargesheet | test_features_4_to_10.py |
| 5 | Witness Interview Summarizer | Frontline UX | NLP distillation of 5-page testimonies into 3 bullets | test_witness_summarizer.py |
| 6 | Court Evidence Zip Bundle | Frontline UX | AES-sealed evidentiary bundle with SHA-256 manifests | test_secure_bundle.py |
| 7 | Instant Cross-Case Linker | Frontline UX | Graph neighborhood discovery of shared burner SIMs | Cytoscape fCoSE verified |
| 8 | Statutory Legal Section Assistant | Frontline UX | BNS-to-IPC concordance mapping for charge-sheets | Legal API mapper verified |
| 9 | Live Highway Intercept Proximity | Frontline UX | Haversine/OSRM police station tactical routing | test_police_proximity.py |
| 10 | Air-Gapped Hash Manifest | Frontline UX | WORM hash seal for physical seized devices & files | HMAC SHA-256 ledger test |
| 11 | Prometheus Metrics & Sentry | AI / MLOps | Full system observability & telemetry tracking | test_advanced_features.py |
| 12 | Model Drift Detection (KS-Test) | AI / MLOps | Automated statistical drift detection for fraud ML | test_feature2_drift.py |
| 13 | Federated Learning Roster | AI / MLOps | Privacy-preserving model aggregation across stations | FedAvg algorithm test |
| 14 | Merkle Root Blockchain Notary | AI / MLOps | Binary Merkle tree proof over immutable WORM ledger | test_feature4_merkle.py |
| 15 | SMS/WhatsApp Field Dispatcher | AI / MLOps | Rate-limited urgent alerts to beat constables | test_feature5_sms.py |
| 16 | Voice Biometrics Identification | AI / MLOps | 256-d voiceprint extraction for wiretap audios | test_feature6_voice.py |
| 17 | Automated Bail Opposition Draft | AI / MLOps | Sec 480 BNSS / 437 CrPC counter-arguments | test_feature7_bail.py |
| 18 | TAXII/STIX Threat Intel Sync | AI / MLOps | Automated ingestion of CERT-In / I4C indicators | test_feature8_taxii.py |
| 19 | Automated Smoke Load Testing | AI / MLOps | High-concurrency SLA verification (P99 < 500ms) | test_feature9_load.py |
| 20 | 3D Crime Scene Photogrammetry | AI / MLOps | Point-cloud (.ply) reconstruction from photos | test_feature10_3d.py |
| 21 | Multi-Tenant Row-Level Security | Operations | Strict station-level data isolation on shared cloud | test_multi_tenant_rls.py |
| 22 | Automated Backup & Disaster Recovery | Operations | Daily encrypted snapshots with SHA-256 verification | test_backup_dr.py |
| 23 | Audit Time-Travel Query Engine | Operations | Historical graph reconstruction at timestamp T | test_time_travel.py |
| 24 | Case Aging & Stalled Alerts | Operations | Enforces 60d POCSO / 90d BNSS charge-sheet deadlines | test_case_aging.py |
| 25 | Multi-Jurisdiction Warrant Tracker | Operations | Cross-state NBW & 15-day expiring LOC tracker | test_warrant_tracker.py |
| 26 | Cold Case Resurrection Engine | Operations | Reverse-matching new suspects to historical untraced FIRs | test_cold_case_engine.py |
| 27 | Officer Workload Balancer | Operations | Caseload severity scoring (Heinous 5pts vs Minor 1pt) | test_workload_balancer.py |
| 28 | Real-Time CDR Streaming Fusion | Operations | Live cell tower handover packets via Redis Streams | test_cdr_live_stream.py |
| 29 | Biometric Hash Matcher (Minutiae) | Operations | DPDP zero-knowledge ISO 19794-2 template matching | test_biometric_matcher.py |
| 30 | SHO Supervisory Morning Briefing | Operations | Daily 08:00 roll call briefing dossier & PDF export | test_sho_briefing.py |

# CRIMENEXUS v3.0 • ARCHITECTURE & DEPLOYMENT BLUEPRINT

**Dual-Database Knowledge Graph · High-Throughput Streaming · Test Scoreboard**

## 1. High-Level Architectural Flow

| Layer | Technology | Operational Police Role |
|---|---|---|
| Data Intake | OCR / PyMuPDF / OpenCV / Regex | Ingestion of Scanned FIRs, Telecom CDRs, Bank Excel, DVR H.264 Footage |
| Streaming Layer | Redis 7.0 Streams (Event Driven) | Sub-millisecond pub/sub queue: `bank.txns`, `telecom.cdr`, `anpr.sightings` |
| Relational Tier | PostgreSQL 15 (ACID) | Cases, Accused metadata, WORM HMAC-SHA256 immutable audit ledger |
| Graph Engine | Neo4j 5.24 Community (APOC) | Deep syndicate traversal: `(:Person)-[:USES_PHONE]->(:Phone)` |
| Inference Tier | Scikit-Learn / Isolation Forest / Ollama | SHAP XAI explainable risk scoring, smurfing anomaly detection, RAG drafting |
| Security / Tenancy | Row-Level Security (RLS) + JWT | Strict station isolation (IO/SHO) with DSP/SP/CID oversight tiers |
| Frontline Client | React 19 + TypeScript + Cytoscape + MapLibre | Standard WebGL workspace + 200 KB Thana Lite Mode for 2 Mbps connections |

## 2. Full Regression Testing Scoreboard (Commit 07bf9c2)

| Test Suite Component | Test Count | Pass Rate | Execution Time | Verdict |
|---|---|---|---|---|
| Full Backend Pytest Suite | 172 Tests | 100% (172/172) | 84.09s | PASSED |
| Dedicated Automation Proofs | 40 Tests | 100% (40/40) | 33.51s | PASSED |
| Advanced Intelligence & Ops Suite | 57 Tests | 100% (57/57) | 20.83s | PASSED |
| Ground Scenario End-to-End Tests | 17 Tests | 100% (17/17) | 24.12s | PASSED |
| Frontend Quality & Unit (Vitest) | 13 Tests | 100% (13/13) | 9.88s | PASSED |
| High-Concurrency Smoke Load Test | 48 Concurrent Req | 0.000% Error | P99: 107.0ms (< 500ms SLA) | PASSED |
| Production Multi-Stage Docker Build | 2 Stages (Builder/Runner) | Zero Pip Errors | 270s (562 MB Image) | VERIFIED |

## 3. Operational Deployment Specifications

**Single-Server Footprint:** 8 CPU Cores · 16 GB RAM · 250 GB NVMe Storage · Ubuntu 22.04 LTS.
**License & Sovereignty:** MIT Open Source License · Zero recurring software royalties · Data remains 100% on state premises.
**Legal Admissibility:** Automated Section 63 BSA / Section 65B IEA Electronic Evidence Certificates generated with SHA-256.